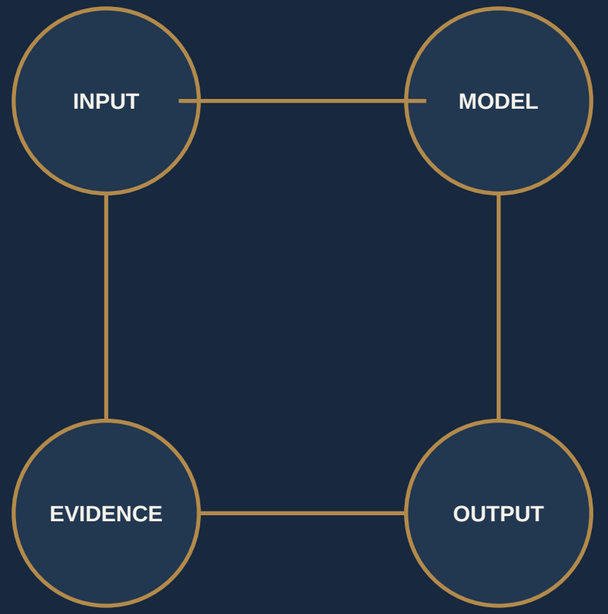
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Operating Systems

AI.304 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define ros/ros 2 concepts.
- Apply key methods including nodes.
- Build or analyze a practical system involving topics.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

ROS/ROS 2 concepts

02

METHODS

Nodes

03

SYSTEMS

Topics

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · ROS/ROS 2 concepts

ROS/ROS 2 concepts

02 · Nodes

Nodes

03 · Topics

Topics

04 · Services

Services

05 · Actions

Actions

06 · Packages

Packages

07 · Launch systems

Launch systems

08 · Sensors and robot simulation

Sensors and robot simulation

Core ideas, made explicit

AI.304 • Distinguish definitions, assumptions and evidence

- ROS/ROS 2 concepts
- Nodes
- Topics
- Services
- Actions

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.304 • Distinguish definitions, assumptions and evidence

- Packages
- Launch systems
- Sensors and robot simulation
- Mandatory practical: Build a complete ROS-based robot application and demonstrate it in simulation

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
ROS/ROS 2 concepts

INPUTS
Nodes

METHOD
Topics

OUTPUT
Services

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving ros/ros 2 concepts.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using ros/ros 2 concepts.
- Implement or simulate a baseline using nodes.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.